



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: User Interface and Experience Design

Course Code: TE7952

Faculty: Engineering

Course Credit: 3

Course Level: 4

Sub-Committee (Specialization): Computer Science

Learning Objectives:

Students will be able to-

1. Understand iterative user-centered design of graphical user interfaces
2. Apply the user Interfaces to different devices and requirements
3. Create high quality professional documents and artifacts related to the design process.

Books Recommended:

Book	Author	Publisher
A Project Guide to UX Design: For user experience designers in the field or in the making	Russ Unger and Carolyn Chandler	New Riders Publishing
The Elements of User Experience: User-Centered Design for the Web and Beyond	Jesse James Garrett	Pearson
The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques	Wilbert O. Galitz	Wiley
The UX Book Process and Guidelines for Ensuring a Quality User Experience	Rex Hartson and Pardha S. Pyla	Elsevier

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Introduction to the UI:What is User Interface Design UI -The Relationship Between UI and UX , Roles in UI/UX, A Brief Historical Overview of Interface Design, Interface Conventions, Approaches to Screen Based UI, Template vs Content, Formal Elements of Interface Design, Active Elements of Interface Design, Composing the Elements of Interface Design, UI Design Process, Visual Communication design component in Interface Design	7	7
2	Introduction to UX:UX Basics- Foundation of UX design, Good and poor design, Understanding Your Users, Designing the Experience, Elements of user Experience, Visual Design Principles, Functional Layout, Interaction design, Introduction to the Interface, Navigation Design, User Testing, Developing and Releasing Your Design	8	8

3	Fundamentals of UI, heuristics and interaction designUI Elements-Patterns, Material Design Google and Human Interface Design Apple guidelines, Interaction Principles Interaction Behaviour, Master the Brand Platforms Style Guides, Understand problems and design solutions for e-commerce, social media, message, data, and dashboard design.Principles of Sketching,Core Responsive Design, Wireframing vs Wireflows, Click through Wireframing Prototyping,Work with different tools Figma, Sketch, Adobe XD	10	10
4	UI/ UX Design Tools:User Study- Interviews, writing personas: user and device personas, User Context, Building Low Fidelity Wireframe and High-Fidelity Polished Wireframe Using wireframing Tools, Creating the working Prototype using Prototyping tools, Sharing and Exporting Design.	12	12
5	Usability Evaluation and Product DesignType of usability evaluation Qualitative Quantitative evaluation.Guerilla testing , A/B Testing, Unmoderated remote usability testing,Card sorting, Designing evaluation protocol, Conducting usability evaluation study,Types of products solutions, Design Psychology for e-commerce sites, Design Thinking Life Cycle, UX practices in corporate World.	8	8
Total		45	45

Pre Requisites:

HTML, CSS, Fundamentals of graphics design, principles of visual design

Evaluation:

A.Continuous Assessment: 1.Unit Tests 2.Assignments 3.Quizzes 4.Class Test
B. End Semester Examination: 1. Written Examination

Pedagogy:

1. Classroom teaching
2. Hands on Lab exercises
3. Case studies
4. Project-based learning

Expert:

Viplav Shah,Design Manager,FanCode